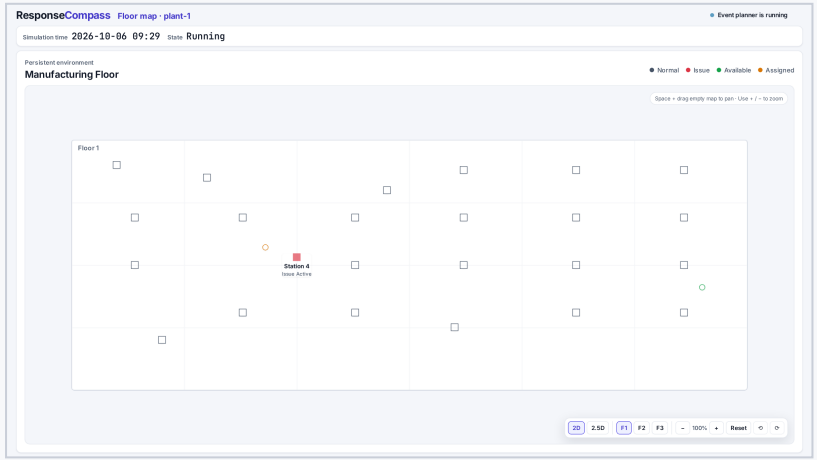
ResponseCompass

The right responder. The right fix.

Learned from every past repair.

Runs autonomously. Supervised by your control room.

An AI dispatch agent for operations control rooms, built on AMD Token Factory.



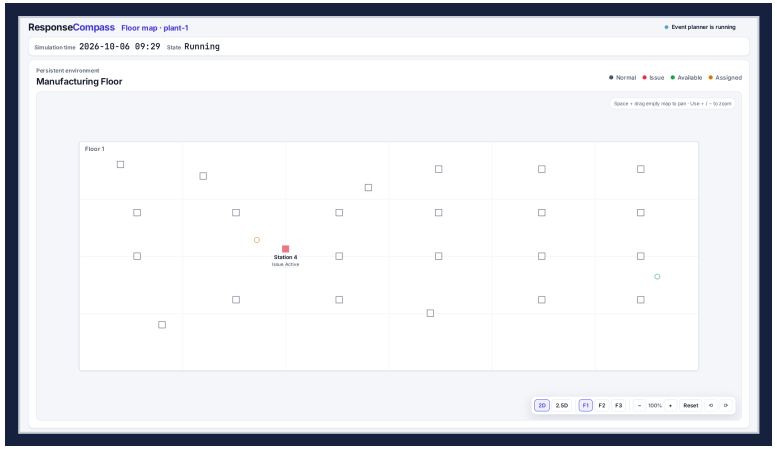
Floor-map wall screen, simulated plant e8-aura-p1

The problem

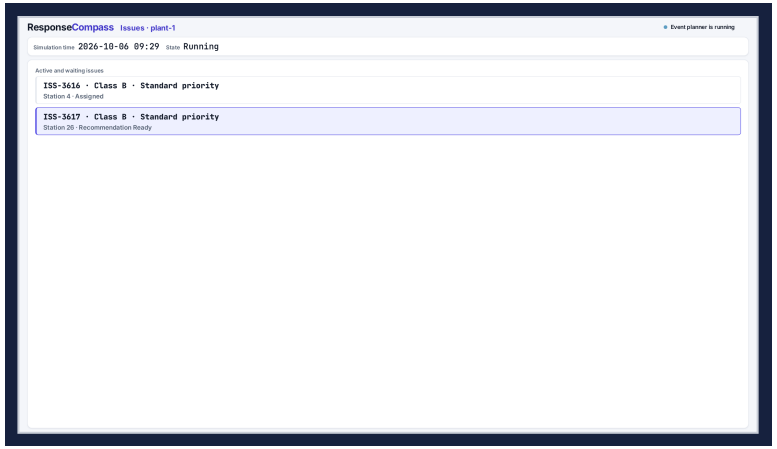
- When equipment fails, **closest** or **free** may be an incomplete answer.
- The nearest technician may never have fixed this kind of fault, and the first fix tried is often a guess.
- A poor match means downtime, a reopened issue or another trip; a backlog means overtime decisions under pressure.
- Who fixes what, and how, lives in people's heads and old tickets.

One agent, many screens

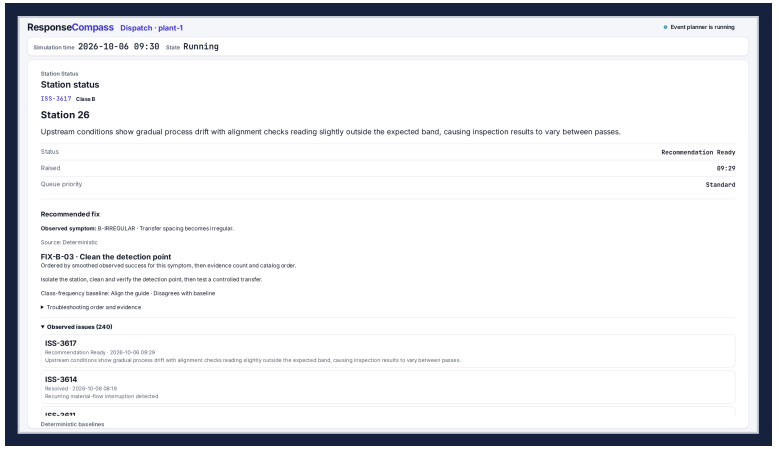
Each panel can run full screen on its own monitor; the operator console keeps the controls.



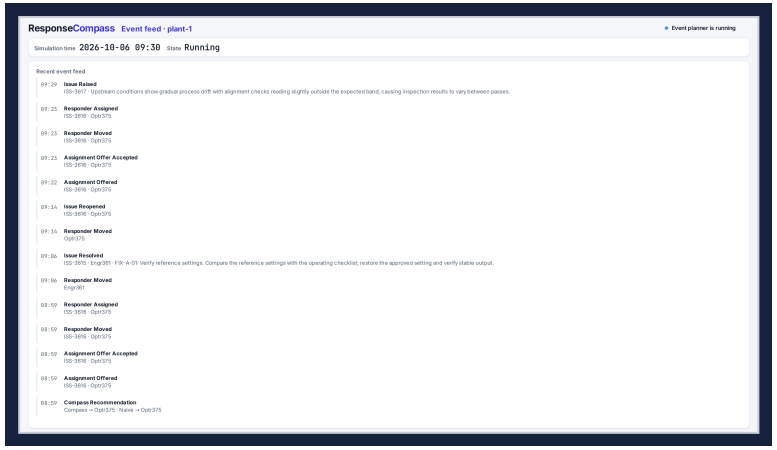
Floor map



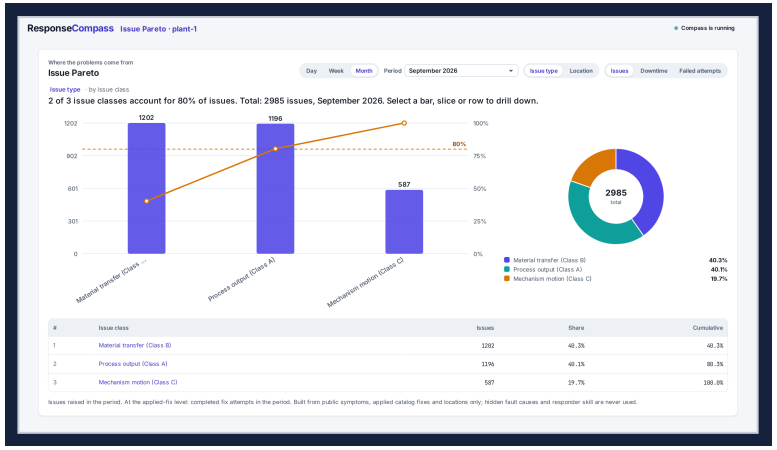
Issues



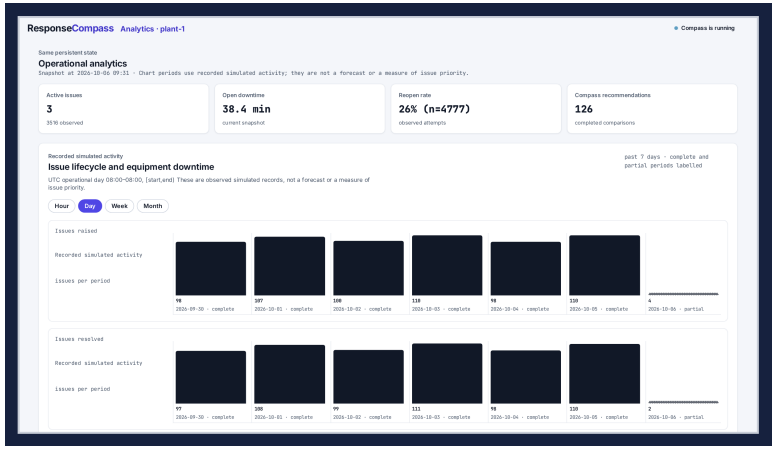
Dispatch



Event feed



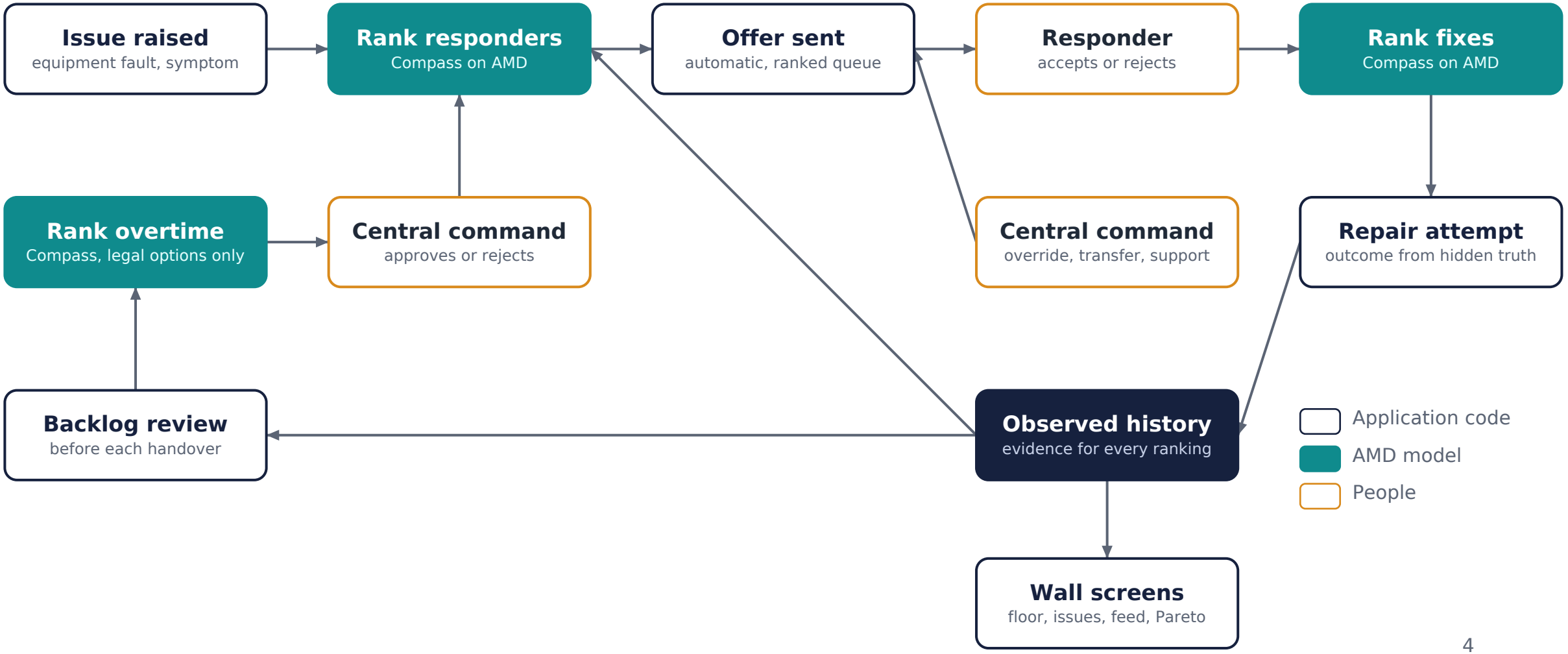
Issue Pareto



Analytics

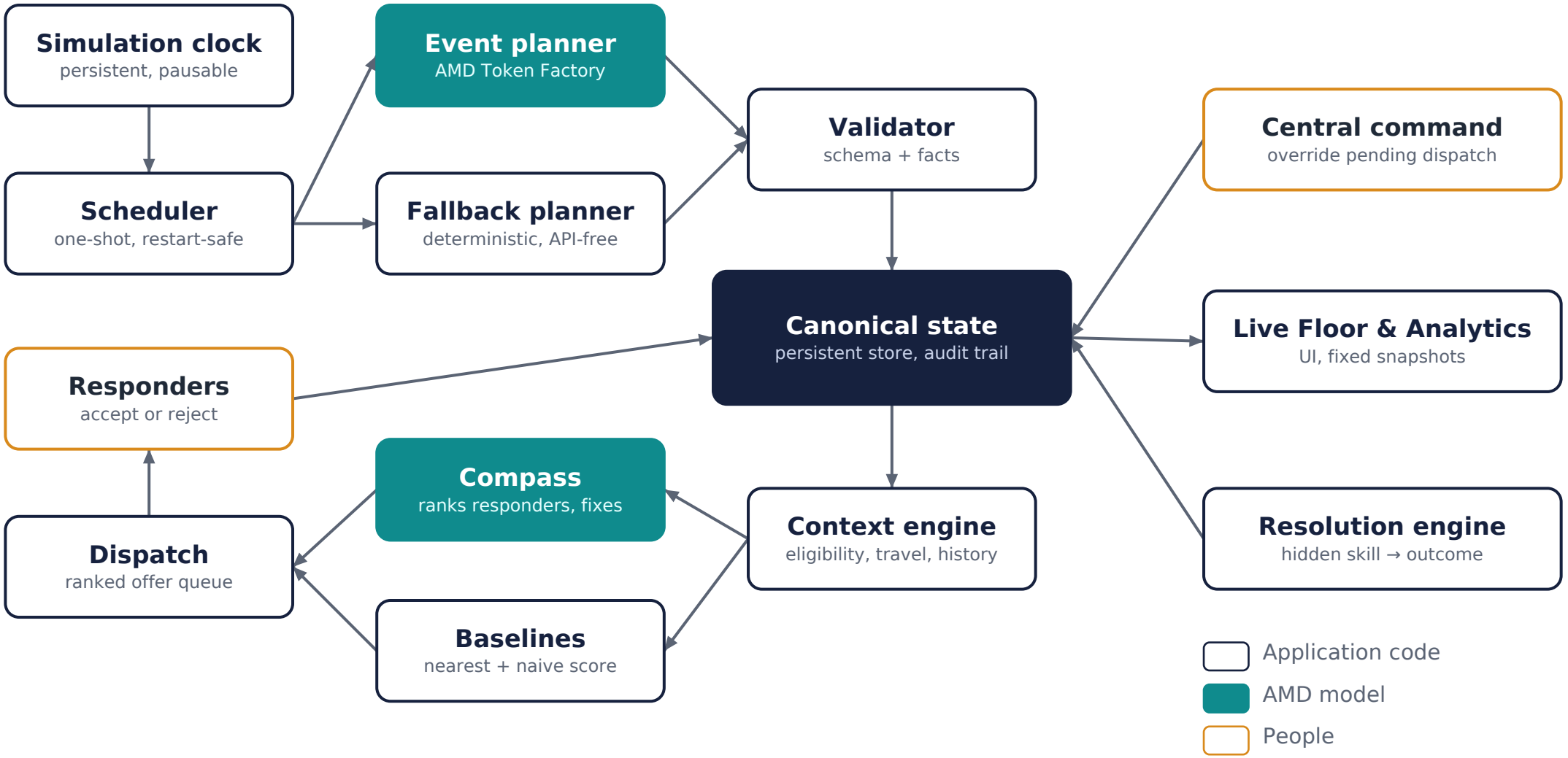
Autonomous by default, human in command

Code runs the loop; AMD models rank; people decide at the amber steps.



Code owns the facts. Models only rank.

Application code owns time, facts, eligibility, baselines and every mutation; models only propose.



Honest by design

- Each responder's **true skill** and each fault's **true cause** are hidden simulation truth: no model, screen or analytics sees them.
- Outcomes are simulated from that hidden truth, independently of what the model claims.
- Transparent baselines run alongside (nearest, naive score, most common fix); disagreements are recorded.
- Analytics show denominators, suppress thin samples and are not an employee leaderboard.
- Excluded from ranking: pay, seniority, demographics, fatigue, HR data.

The right responder

ISS-3618, Station 48 · captured live on AMD Token Factory (MiMo-V2.6-Flash)

- Nearest responder **Engr361**: half a minute away, 26 similar cases.
- Compass ranked **Engr370** first: 110 similar cases, 87% observed success, the same half-minute away, and explained why.
- Offer accepted automatically; central command could have overridden it at any time.

Initial assignment Compass analysis

Engr370 was ranked first.

Engr370 combines the deepest issue-specific history, the most reliable evidence, and the best observed balance of success and limited reopens, while still requiring only a modest trip relative to the alternatives.

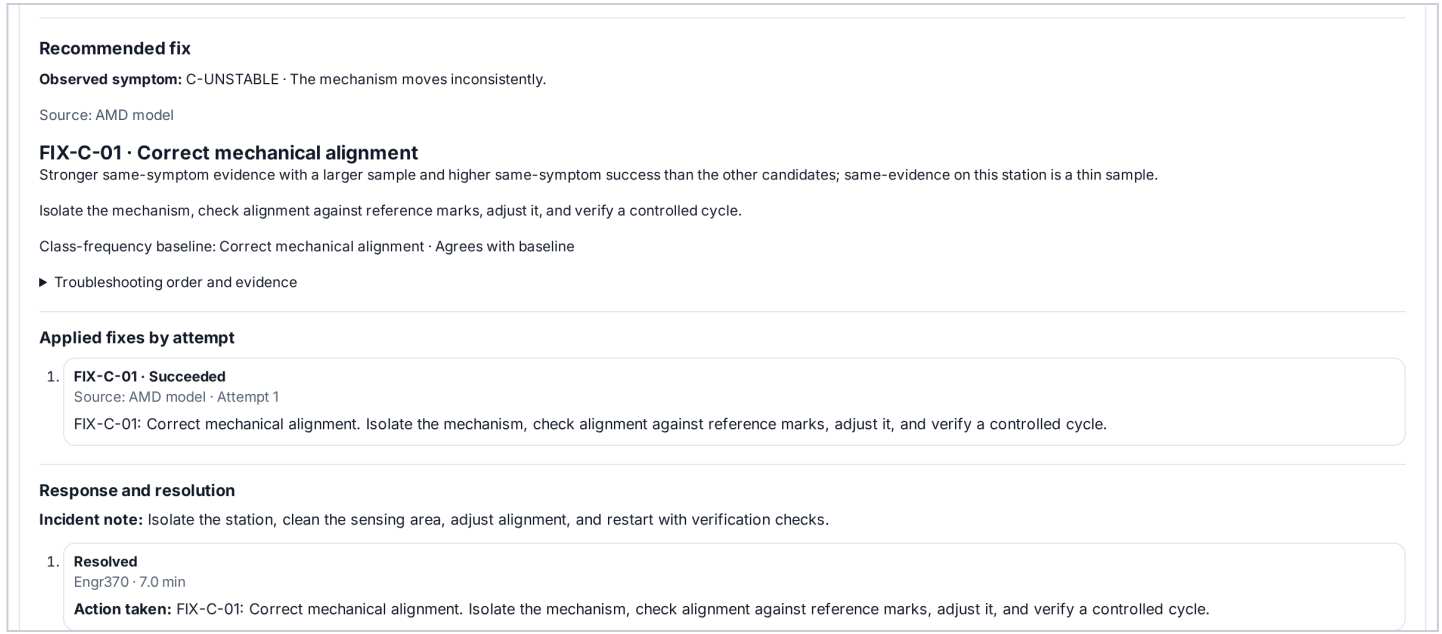
Eligible responder evidence

Candidate	Solo / assisted	Evidence weight	Strength	Median	Success	Travel
1. Engr370	17 / 93	62.9	STRONG	8.8 min	87% (n=62.9)	0.5 min
2. Engr361	3 / 23	10.4	STRONG	13.9 min	83% (n=10.4)	0.5 min
3. Tech374	0 / 7	4.1	MODERATE	12.1 min	66% (n=4.1)	1.8 min

The right fix

Same record: Compass ranks the troubleshooting procedures for the observed symptom

- Code supplies the catalog procedures and their **observed** success on this symptom, class and station.
- Compass ranked **Correct mechanical alignment** first; the most common fix is the baseline.
- Engr370 applied it: fixed **first time in 7.0 min**.
- A failed attempt reopens the issue and the next procedure is tried.



Recommended fix

Observed symptom: C-UNSTABLE · The mechanism moves inconsistently.

Source: AMD model

FIX-C-01 · Correct mechanical alignment

Stronger same-symptom evidence with a larger sample and higher same-symptom success than the other candidates; same-evidence on this station is a thin sample.

Isolate the mechanism, check alignment against reference marks, adjust it, and verify a controlled cycle.

Class-frequency baseline: Correct mechanical alignment · Agrees with baseline

▸ Troubleshooting order and evidence

Applied fixes by attempt

1. **FIX-C-01 · Succeeded**
   Source: AMD model · Attempt 1
   FIX-C-01: Correct mechanical alignment. Isolate the mechanism, check alignment against reference marks, adjust it, and verify a controlled cycle.

Response and resolution

Incident note: Isolate the station, clean the sensing area, adjust alignment, and restart with verification checks.

1. **Resolved**
   Engr370 · 7.0 min
   Action taken: FIX-C-01: Correct mechanical alignment. Isolate the mechanism, check alignment against reference marks, adjust it, and verify a controlled cycle.

Overtime, decided by people

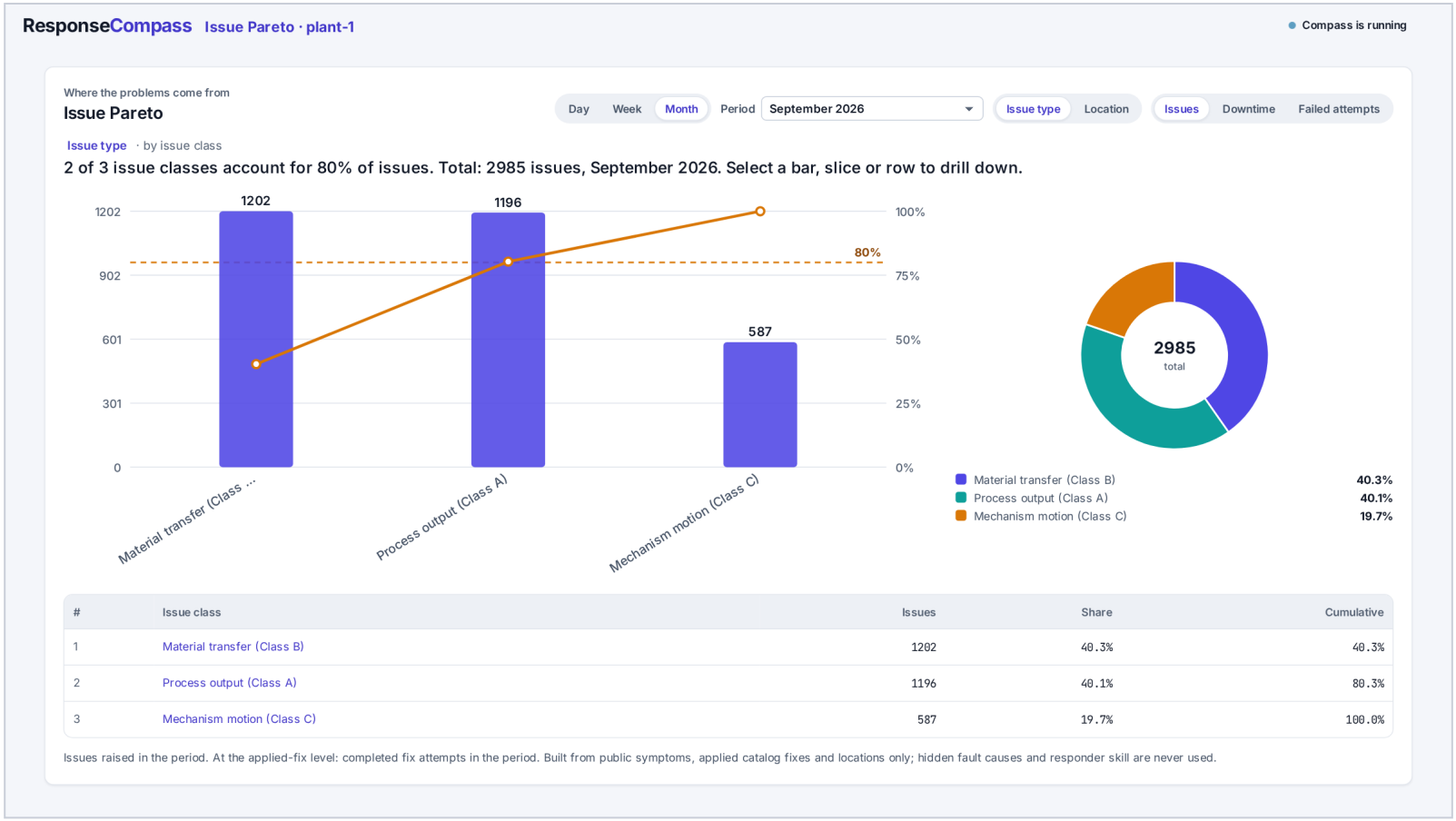
Backlog review → lawful options → Compass ranking → central command approves or rejects

- When the backlog outgrows the shift, code lists only **legal** options: 16 h duty, 12 h rest, staffing cap.
- Compass ranks them by observed coverage of the backlog; the old first-eligible rule is the baseline.
- Nothing changes until **central command** approves; unanswered recommendations expire.
- Every decision, and who made it, is audited.



Where the downtime comes from

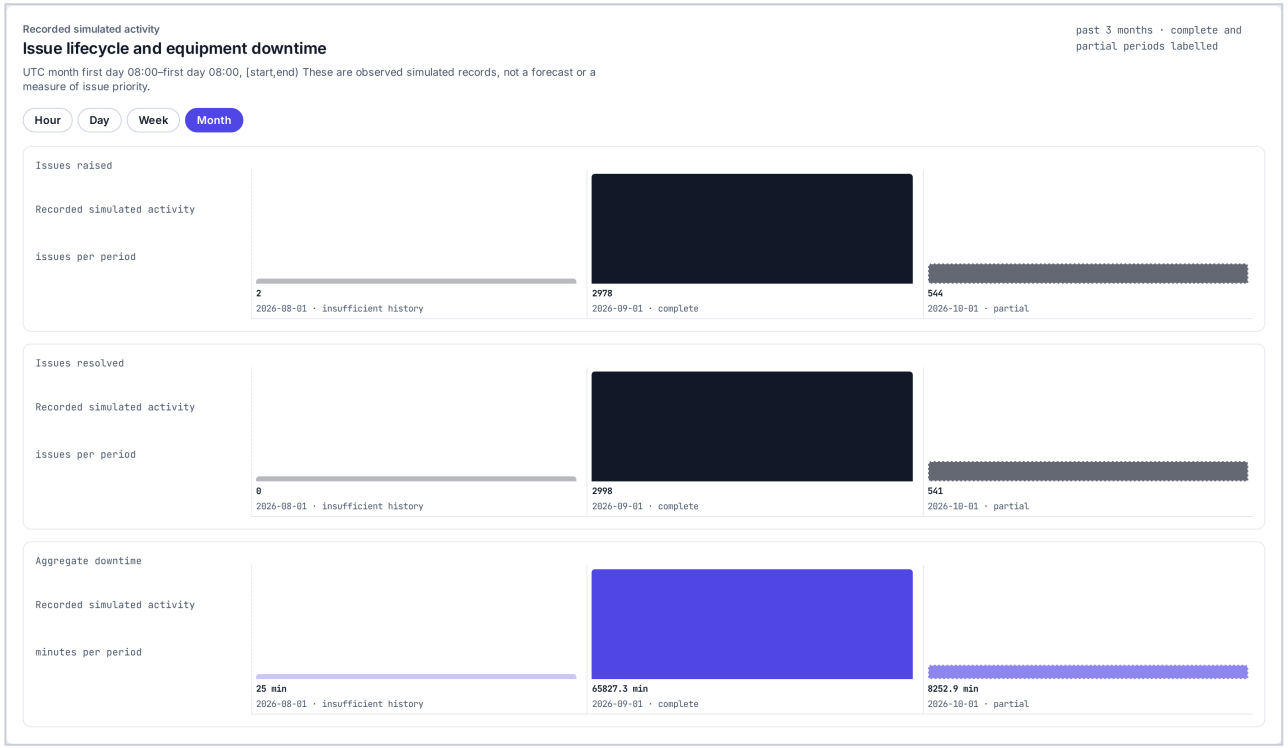
Issue Pareto: issue class → observed symptom → applied fix, or floor → station



- Issues, downtime or failed attempts per day, week or month.
- Sorted bars, cumulative line and an **80%** marker, plus a donut.
- Click to drill from class to symptom to the fix that was applied.
- Built from public records only; hidden causes are never used.

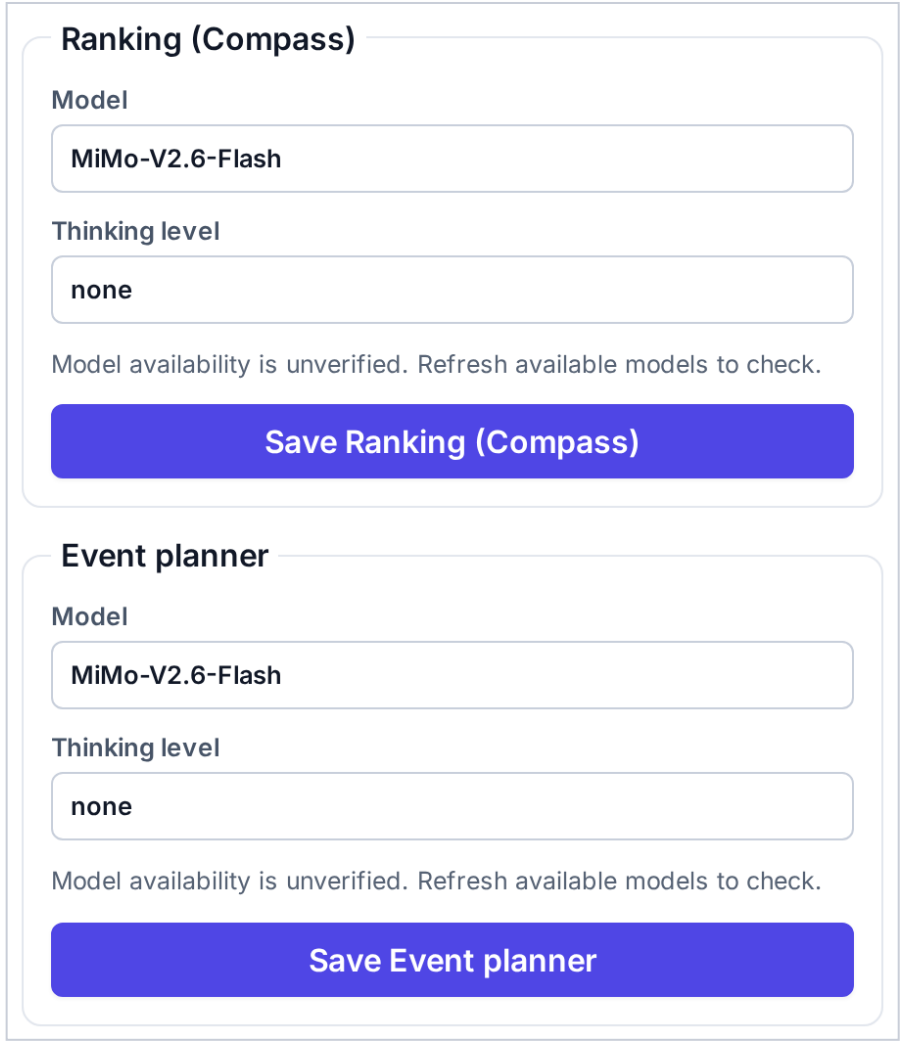
A month of operations

- e8-aura-p1: **3,512 issues**, 1 September to 6 October (simulated).
- September: **2,966 raised / 2,985 resolved** events.
- First-time fix rate in September: **73.5%** (1,858 of 2,527 first attempts).
- Weekly first-time fix: 67.1%, 74.9%, 76.5%, 75.6%: not a steady climb.



Built on AMD Token Factory

- OpenAI-compatible API, JSON output; our validators are the contract.
- Default model **MiMo-V2.6-Flash** ranks responders, fixes and overtime; the event planner can use another model.
- Local budget: 10 requests/min, 2 concurrent; bounded 429 retries; circuit breaker.
- Recorded MiMo check: **9/10** valid responder rankings, **10/10** valid event plans.
- Model calls run in a background lane and never hold the simulation clock.



Engineering quality

- Live and Offline are one app; Offline makes no API calls and is fully reproducible.
- **317** hermetic unit and integration tests and **67** Chromium end-to-end tests.
- Deterministic seeded simulation; restart-safe scheduler; atomic, crash-safe persistence.
- Runs on Node.js 22 or Docker; no external services besides the model API.

What's next

- Connect to real maintenance, ticketing and rostering systems in place of the simulated floor.
- Learn from approvals: feed operator decisions and repair outcomes back into the evidence.
- Measure Compass against its baselines on live AMD rankings over a longer run.
- Long-run storage: compact the remaining growing history for year-long runs.

The right responder. The right fix.